mitochondrial membrane is impermeable to small molecules, with the exception of oxygen, carbon-dioxide and water. All of the other substrates of mitochondrial metabolism, as well as its products, therefore have to be moved through the innermembrane with the help of special transporters. Embedded within the innermembrane of the mitochondria are various protein carriers, principally cytochromes.

The respiratory chain is organized in the form of multi-protein complexes. The inner space also contains ATP synthase and other enzymes. The inner membrane plays a major role in oxidative phosphorylation. The matrix of the mitochondrion is like a gel with ~50% protein. The matrix contains enzymes involved in the oxidation of pyruvate, amino acids, fatty acids, TCA cycle, as well as NAD^+, FAD, ADP and Pi.

The energy liberated during the oxidation of carbohydrate and food is available in mitochondria as reducing equivalents (2H) that are collected by the respiratory chain for oxidation and coupled generation of ATP (Figure 8.2).

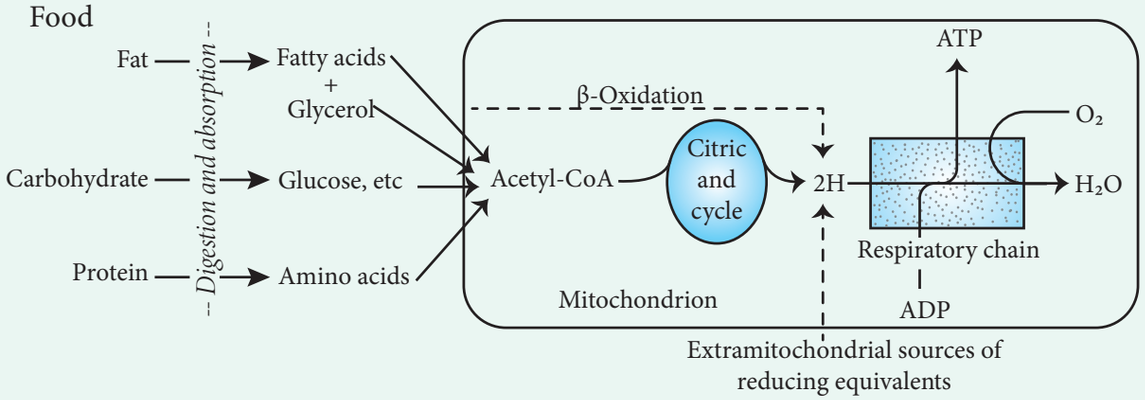


Figure 8.2. Digestion and absorption of food and production of energy

8.2.2 Components of the electron transport chain

The components of the ETC in eukaryotes are located in the inner mitochondrial membrane and organized into four large protein complexes (Figure 8.3). These complexes consist of several proteins and prosthetic groups. Apart from these, two other molecules, coenzyme Q and cytochrome c also play a significant role in the ETC. Complex I, also known as NADH-coenzyme Q oxidoreductase, or NADH dehydrogenase, catalyses the transfer of electrons from NADH to Coenzyme Q (CoQ or Ubiquinone UQ), because it is ubiquitous in biological systems). It contains one molecule of flavin mononucleotide (FMN) and 8 iron-sulfur clusters that participate in electron transfer. Complex II [Succinate Dehydrogenase or Succinate–Coenzyme Q Oxidoreductase] contains succinate dehydrogenase and three small hydrophobic subunits, and pass electrons from succinate to CoQ. Complex III is Coenzyme QH_2-cytochrome c oxidoreductase. It passes electrons from reduced CoQ to cytochrome c. It contains cytochromes b, c1 and one [2Fe-2S] cluster. Complex IV is cytochrome c oxidase. It catalyses the one-electron oxidation of four consecutive reduced cytochrome c molecules and four-electron reduction of

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